Abstract Algebra I (MAT5CJ302): November 2026 paper traced to the textbook

Paper: Fifth Semester CUFYUGP, November 2026, 2024 admission onwards (code D153394 / 795780). Book: Fraleigh, *A First Course in Abstract Algebra*, 8th ed. (the copy on disk; the syllabus names the 7th ed., matched by content). Page references: printed page, with the PDF page of the full 8th-ed book in brackets (PDF = printed + 17). "Unit PDF" means the cut in `SSC-V6/STUDY-PLAN-V5/sources/sem5/maths/abstract-algebra-1/`.

Summary

- **9 of the 22 question parts (Q1–Q18, 19a, 19b, 20a, 20b) are textbook exercises, copied word for word:** Q2 (Ex 2.44), Q12 (Ex 2.29), Q13 (Ex 5.60), Q14 (Ex 6.60), Q15 (Ex 8.48), Q16 (Ex 10.39), Q17 (Ex 8.39), Q19a (Ex 2.37), Q20b (Ex 6.56).
- **8 are theorems, definitions or proofs printed in the text:** Q4 (Def 5.22, Cor 6.17), Q7 (Thm 10.7), Q9 (Def 23.7, Cor 23.5), Q10 (Thm 23.11), Q11 (group-table text, Thm 2.17), Q18 (Thm 23.3), Q19b (the proof just before Def 12.19), Q20a (Thm 4.8).
- **5 are computations of the same kind as a book example or exercise, with different data:** Q1 (Ex 1.8), Q3 (Ex 5.1–5.6), Q5 (Ex 8.17–8.19), Q6 (Ex 10.1–10.5), Q8 (Ex 8.11).
- **Where you should have looked: the "Theory" exercises at the end of §2, §5, §6, §8 and §10.** Section B (36 marks) and both Section C questions come almost entirely from these lists.
- **Gap in the unit PDFs:** the cuts for §5, §6 and §8 stop before the exercise pages (§5 exercises printed 57–60, §6 exercises printed 68–70, §8 exercises printed 85–87). Q13, Q14, Q15, Q17 and Q20b, and the computation models for Q3, Q5 and Q8, are on those missing pages. Q19b is in §12 (Factor Groups), which no unit PDF covers. (The project excerpt `SEM5-PYQ/ABSTRACT-ALGEBRA/source/books/Fraleigh-8ed_Module2_excerpt.html` does include the §5, §6 and §8 exercise pages, printed 52–70 and 77–88.)

Match key: **Exact** same wording as the book; Theory a theorem, definition or proof printed in the text; Same type a book example or exercise with different data.

Section A (3 marks each, ceiling 24)

Q	Question	Where it is in the book	Match	In unit PDF?	Key point
1	$a*b = ab + 1$ on $\mathbb{Q}$: commutative? associative?	§1 Def 1.11 (commutative) and Def 1.12 (associative), p. 13 [30]. Exercises 1.7–1.11, p. 17 [34]; nearest is **Ex 1.8** ($a*b = 2ab + 3$ on $\mathbb{Q}$).	Same type	Yes: unit-1-4	Commutative: yes. Associative: no, $(a*b)*c = abc + c + 1$ but $a*(b*c) = abc + a + 1$.
2	If φ is an isomorphism, then φ^{-1} is an isomorphism.	§2 Def 2.22 (group isomorphism), p. 26 [43]; the text there states the result. **Exercise 2.44**, p. 31 [48]. (The paper's $\langle S,* \rangle$ wording is the 7th-ed §3 binary-structure version.)	**Exact**	Yes: unit-5-6	φ^{-1} is a bijection; put $a = \varphi(x)$, $b = \varphi(y)$ and apply φ^{-1} to $\varphi(x*y) = a*'b$.
3	Is $H = \{(x,y) : x + y = 0\}$ a subgroup of $(\mathbb{R}^2, +)$?	§5 Example 5.3 (a subgroup of $\mathbb{R}^n$), p. 53 [70]; subgroup tests Theorem 5.12 and Theorem 5.15, p. 54 [71]. Exercises 5.1–5.6 (subset tests), p. 57 [74].	Same type	Theory yes (unit-7-8); **exercises no**	Yes. Closed; $(0,0) \in H$; the inverse of (x,y) is $(-x,-y)$, not $(-x,y)$ as in your margin note.

4	Define cyclic group and generator; example with all generators.	§5 Def 5.20 (cyclic subgroup) and **Def 5.22** (generator, cyclic group), Examples 5.23–5.24 ($\mathbb{Z}_4$: 1, 3; $\mathbb{Z}$: ±1), p. 56 [73]. §6 Example 6.16 ($\mathbb{Z}_{12}$: 1, 5, 7, 11) and **Corollary 6.17**, p. 66 [83]. Exercise 5.26, p. 58 [75].	Theory	Yes: unit-7-8, unit-9-10	Generators of $\mathbb{Z}_n$ are the r with gcd(r, n) = 1.
5	Orbits of σ = (1 2 3 4 5 6 7 8 / 5 6 2 8 1 4 7 3).	§8 definition of orbit and Example 8.18, p. 82 [99]. **Exercises 8.17–8.19** (find all orbits; Ex 8.18 is an 8-letter permutation in the same layout), p. 85 [102].	Same type	Definition yes (unit-11-12); **exercises no**	{1,5}, {2,6,4,8,3}, {7}.
6	All cosets of H = {0,4} in $\mathbb{Z}_8$.	§10 Def 10.2 and **Example 10.4** (cosets of ⟨3⟩ in $\mathbb{Z}_{12}$), p. 98 [115]. Exercises 10.1–10.5 (find all cosets), p. 102 [119].	Same type	Yes: unit-15-16	H, 1+H = {1,5}, 2+H = {2,6}, 3+H = {3,7}.
7	|G| = 24: possible orders of a subgroup.	§10 **Theorem 10.7** (Lagrange), p. 99 [116]. Cyclic case: Corollary 6.20 and Example 6.21 (orders of subgroups of $\mathbb{Z}_{28}$), p. 67 [84].	Theory	Yes: unit-15-16	1, 2, 3, 4, 6, 8, 12, 24.
8	φ: $\mathbb{Z} \to \mathbb{Z}_{10}$, φ(n) = 6n: ker φ and φ(18).	§8 **Def 8.6** (kernel), p. 78 [95]. **Exercise 8.11** (φ: $\mathbb{Z} \to \mathbb{Z}_8$ with φ(1) = 6; compute the kernel), p. 85 [102].	Same type	Definition yes (unit-11-12); **exercises no**	ker φ = 5$\mathbb{Z}$ ($10 \mid 6n \Leftrightarrow 5 \mid n$). φ(18) = 108 mod 10 = 8.
9	Define integral domain; $\mathbb{Z}_p$ is one for p prime.	§23 **Def 23.7** and Example 23.8, p. 196 [213]. **Corollary 23.5** ($\mathbb{Z}_p$ is a field with no divisors of 0), p. 195 [212].	Theory	Yes: unit-21-22	$ab \equiv 0 \pmod p \Rightarrow p \mid a$ or $p \mid b$.
10	Every finite integral domain is a field.	§23 **Theorem 23.11** with proof, p. 197 [214].	Theory	Yes: unit-21-22	$x \mapsto ax$ is one-to-one, so onto; solve $ax = 1$.

Section B (6 marks each, ceiling 36)

Q	Question	Where it is in the book	Match	In unit PDF?	Key point
11	Table with identity, each element once in every row and column: a∗x = b and y∗a = b have unique solutions.	§2 "Properties of Group Tables", p. 27 [44] (the converse argument is stated there); Theorem 2.17, p. 24 [41]. Not among the 8th-ed exercises; the wording is the 7th-ed version.	Theory	Yes: unit-5-6	b occurs exactly once in row a, so exactly one column x gives a∗x = b; columns for y.
12	|G| even ⇒ some a ≠ e with a∗a = e.	**Exercise 2.29**, p. 31 [48]. (Ex 10.38, p. 104 [121], refers back to it.)	**Exact**	Yes: unit-5-6	Pair each element with its inverse; e is alone, so another self-inverse element remains.
13	a ~ b ⇔ $ab^{-1} \in H$ is an equivalence relation.	**Exercise 5.60**, p. 60 [77]. Left-coset version: Theorem 10.1 ($a^{-1}b \in H$), p. 97 [114]; the right relation ~R is named on p. 100 [117].	**Exact**	**Ex 5.60 no**; Thm 10.1 yes (unit-15-16)	Reflexive: $aa^{-1} = e$. Symmetric: $(ab^{-1})^{-1} = ba^{-1}$. Transitive: $(ab^{-1})(bc^{-1}) = ac^{-1}$.

14	a is the only element generating a cyclic subgroup of order 2 ⇒ a is the unique element of order 2.	**Exercise 6.60**, p. 70 [87].	**Exact**	**No**	b of order 2 ⇔ ⟨b⟩ has order 2; so b = a.
15	σ a cycle of odd length ⇒ σ² is a cycle.	**Exercise 8.48**, p. 87 [104] (Ex 8.49 generalises it).	**Exact**	**No**	For σ = $(a_1,\ldots,a_n)$, n = 2k+1: σ² = $(a_1,a_3,\ldots,a_{2k+1},a_2,a_4,\ldots,a_{2k})$.
16	At least two elements, no proper nontrivial subgroups ⇒ finite of prime order.	**Exercise 10.39**, p. 104 [121]. Uses Ex 5.64 (such a group is cyclic), p. 60 [77], and Theorem 6.6, p. 63 [80].	**Exact**	Yes: unit-15-16 (last page)	G = ⟨a⟩; if infinite, $\langle a^2\rangle$ is proper; if |G| = mn, $\langle a^m\rangle$ is proper.
17	Composite of homomorphisms γφ is a homomorphism.	**Exercise 8.39**, p. 87 [104]. Homomorphism Def 8.1, p. 77 [94].	**Exact**	**Ex 8.39 no**; Def 8.1 yes	γφ(ab) = γ(φ(a)φ(b)) = γφ(a)·γφ(b).
18	In $\mathbb{Z}_n$ the divisors of 0 are the nonzero elements not relatively prime to n.	§23 Def 23.2 (divisor of 0), p. 194 [211]; **Theorem 23.3**, p. 195 [212]. The paper's wording is the 7th-ed statement (Theorem 19.3 there); the 8th ed. combines it with the units case.	Theory	Yes: unit-21-22	gcd(m,n) = d > 1: m(n/d) = 0. gcd = 1: ms = 0 ⇒ n | s ⇒ s = 0.

Section C (answer one, 10 marks)

Q	Question	Where it is in the book	Match	In unit PDF?	Key point
19a	$(a*b)^2 = a^2*b^2 \Rightarrow a*b = b*a$.	**Exercise 2.37**, p. 31 [48].	**Exact**	Yes: unit-5-6	abab = aabb; cancel a on the left and b on the right.
19b	i_g(x) = gxg′ is an isomorphism of G with itself.	§12 (Factor Groups), the paragraph just before **Def 12.19** (inner automorphism), p. 118 [135]. The full proof is printed there.	Theory	**No (§12 is Module V, not cut)**	Homomorphism: gxyg′ = gxg′·gyg′. One-to-one by cancellation. Onto: y = i_g(g′yg).
20a	S_A is a group under permutation multiplication.	§4 Def 4.6 (permutation), p. 41 [58]; **Theorem 4.8** with proof, p. 43 [60].	Theory	Yes: unit-13-14 (Module 3 file)	Closure (composite of bijections), associativity of composition (Thm 1.13), identity ι, inverse σ^{-1}.
20b	ab has finite order n ⇒ ba has order n.	**Exercise 6.56**, p. 69 [86].	**Exact**	**No**	$(ba)^k = a^{-1}(ab)^k a$; so $(ba)^k = e \Leftrightarrow (ab)^k = e$.

Where to look next time

Book section	Pages (printed [PDF])	Questions on this paper	In unit PDFs?
§2 Exercises 28–45 (Theory)	31 [48]	Q2, Q12, Q19a	Yes

§5 Exercises 1–6, 26, 60, 64	57–60 [74–77]	Q3 (type), Q13, model for Q16	**No**
§6 Exercises 56, 60	69–70 [86–87]	Q14, Q20b	**No**
§8 Exercises 11, 17–19, 39, 48	85–87 [102–104]	Q5 (type), Q8 (type), Q15, Q17	**No**
§10 Exercises 1–5, 38–39	102–104 [119–121]	Q6 (type), Q16	Yes
§12 text before Def 12.19	118 [135]	Q19b	**No**
§23 Def 23.2, 23.7; Thm 23.3, Cor 23.5, Thm 23.11	194–197 [211–214]	Q9, Q10, Q18	Yes
§4 Thm 4.8; §10 Thm 10.7; §5 Def 5.22; Cor 6.17	43, 99, 56, 66	Q20a, Q7, Q4	Yes

Fix for the study set: add the exercise pages of §5 (printed 57–60), §6 (68–70) and §8 (85–87) to the unit PDFs, and work through every "Theory" exercise in §2, §5, §6, §8 and §10. These five lists gave 9 of the 20 questions word for word.